THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — SCIENCE · MID-TERM PRACTICE PAPER

Session 2026-27 · Term I · Chapters 1–7

Class: 8	Subject: Science	Set 02	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. Anita collects a drop of pond water to look for tiny living organisms. Which instrument must she use? — 1

(a) Binoculars (b) Telescope
(c) Magnifying glass only (d) Compound microscope

II. Which cell organelle is present in plant cells but absent in animal cells? — 1

(a) Nucleus (b) Mitochondria (c) Chloroplast (d) Cytoplasm

III. A child eating only polished rice develops swollen gums that bleed. The deficiency is most likely of: — 1

(a) Calcium (b) Vitamin A (c) Vitamin C (d) Iron

IV. Which of the following is a non-communicable disease? — 1

(a) Malaria (b) Typhoid (c) Dengue (d) Diabetes

V. The energy conversion taking place in an electric heater is: — 1

(a) chemical energy to light energy (b) heat energy to electrical energy
(c) electrical energy to heat energy (d) electrical energy to sound energy

VI. The magnetic field around a straight current-carrying wire can be detected using: — 1

(a) a thermometer (b) a magnetic compass
(c) a voltmeter only (d) a lens

VII. Two equal forces act on a body in opposite directions along the same line. The net force on the body is: — 1

(a) double of one force (b) equal to one force
(c) zero (d) cannot be determined

VIII. Which of these forces acts on a body without any contact? — 1

(a) Muscular force (b) Frictional force
(c) Force of a push (d) Gravitational force

IX A drinking straw lets us sip a cold drink because: 1

(a) atmospheric pressure pushes the liquid up when pressure inside the straw is reduced
(b) the straw pulls the liquid
(c) liquid is lighter than air
(d) liquid has no weight

X A tropical cyclone has at its centre: 1

(a) a mountain
(b) a calm region of very low pressure called the eye
(c) strong sunshine and no winds
(d) a region of very high pressure

XI Gases can be compressed easily because: 1

(a) they have fixed shape
(b) their particles are very heavy
(c) their particles attract strongly
(d) there is a large space between their particles

XII When ice melts to form water, the particles: 1

(a) become heavier
(b) gain energy and the arrangement becomes less rigid
(c) lose all their energy
(d) stop moving

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. State any two functions of the cell membrane. **2**

OR

Write any two differences between a plant cell and an animal cell.

3. Mention two benefits of vaccination. **2**

4. What is the heating effect of electric current? Name one appliance based on it. **2**

5. Camels have broad, flat feet. How does this help them walk on soft desert sand? **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

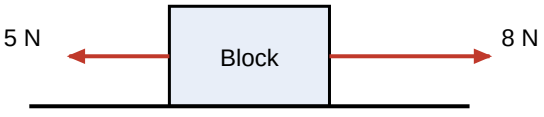
6. Name the disease caused by the dengue virus, state any two symptoms and one preventive measure. **3**

OR

Describe three habits of personal hygiene that help us stay healthy.

7. Explain why the filament of an electric bulb glows while the copper connecting wires do not. **3**

8. Two forces of 8 N and 5 N act on a block along the same straight line in opposite directions as shown. Find the net force and its direction. Is the force balanced? **3**



OR

What are balanced and unbalanced forces? Give one effect of an unbalanced force.

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

9. Complete the given sentences (choose from the words in the box): **(Nucleus, Chloroplast, Cell wall, Cyclone, Fuse)** **5**

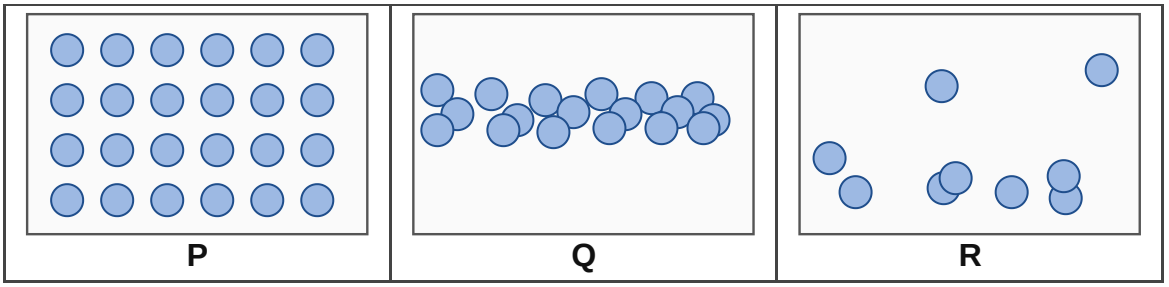
i. The _____ is the control centre of the cell.
ii. _____ contains chlorophyll and traps sunlight for photosynthesis.
iii. A _____ is a safety device that melts and breaks the circuit when excess current flows.
iv. A _____ is a system of rapidly rotating winds around a low-pressure centre.
v. The _____ gives a plant cell its rigid shape.

10. Match the Columns **5**

Column I	Column II
(a) Zinc container	(i) negative terminal of a dry cell
(b) Lactobacillus	(ii) bacterium that turns milk into curd
(c) Newton	(iii) SI unit of force
(d) Gas	(iv) no definite shape and no definite volume
(e) Pascal	(v) SI unit of pressure
	(vi) force of attraction towards the Earth

11. The figure shows the particle arrangement of three states of matter P, Q and R. **5**

(a) Identify P, Q and R and state which has the maximum inter-particle space. (3)
(b) Which one can be compressed easily and why? (2)



OR

(a) Explain how a drinking straw works. (3)
(b) Why do we not feel the huge atmospheric pressure acting on our body? (2)

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. Priya, a Class VIII student, often has bleeding gums and pain in the joints. The doctor advised her to include citrus fruits, tomatoes and green vegetables in her diet.

(a) Deficiency of which vitamin is the likely cause? 1
(b) Name the deficiency disease caused by this vitamin deficiency. 1
(c) Name two food items rich in this vitamin. 2

OR

(d) Why is a balanced diet important for adolescents?

13. After Rohan plugged a heater, an iron and a geyser into the same socket, the electric fuse of his house blew out and all lights went off.

(a) Why did the fuse wire melt? 1
(b) What is this situation of too many appliances in one socket called? 1

(c) State two precautions to avoid such accidents. 2

OR

(d) State one difference between a fuse and an MCB.

14. In May 2019, Cyclone Fani crossed the coast of Odisha with winds of about 175 km/h. Early warnings from the weather department helped authorities shift lakhs of people to cyclone shelters in time.

(a) Over which type of region do tropical cyclones form? 1

(b) What is the central calm part of a cyclone called? 1

(c) Why do cyclones weaken after reaching land? 2

OR

(d) How do early warnings reduce loss of life?

15. In a tug-of-war, both teams pull the rope with a force of 500 N in opposite directions and the rope does not move.

(a) What is the net force on the rope? 1

(b) Are the forces balanced or unbalanced? 1

(c) What will happen to the rope if one team pulls with 600 N and the other with 500 N? Find the net force. 2

OR

(d) Give one example of balanced forces in daily life.